

altairsim — CDOS 2.58 on a Cromemco 16FDC

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CDOS was the operating system of the Cromemco Z80 S-100 machines. It is a work-alike of Digital Research CP/M 1.4 with Cromemco's own extensions, and it runs many CP/M `.COM` programs unchanged. This example boots CDOS 2.58 from an 8" diskette through a **Cromemco 16FDC** floppy controller. One board holds the disk controller, the console UART and the boot PROM.

```
cd examples/cdos
altairsim cdos.toml
```

The machine has a 4 MHz Z80 processor, 64K of RAM and one 16FDC. The CDOS master diskette is in drive A. The machine file sets `bootstrap = true`, which arms the boot PROM. The machine then loads CDOS with no other setting.

Boot the machine

The console shows this text:

```
Preparing to boot, ESC to abort
Standby
CDOS version 02.58
Cromemco Disk Operating System
Copyright (C) 1977, 1983 Cromemco, Inc.

A.
```

CDOS comes up to its `A.` prompt. You do not press a key. The 16FDC is set for a fixed 300-baud modem console (switch 5 of the board). This setting tells the boot PROM to skip the terminal speed measurement. On real hardware, the measurement needed a RETURN from the terminal.

The `A.` prompt shows that CDOS waits on drive A. CP/M shows `A>` for the same state. Type these commands at the prompt:

```
DIR      list the files on the disk
STAT     show the system status: memory, devices, and the disk label and date
```

`DIR` lists the 18 files on the master diskette. They include the assembler (`ASMB`, `LINK`), the editor (`EDIT`), the utilities (`XFER`, `INIT`, `DUMP`, `STAT`), `XMODEM` and CDOS itself. `STAT` reads the label sector of the disk. It prints `CDOS2.58` and the date `05-19-82`.

How the boot works

A reset arms the **RDOS 2.52** boot PROM in the 16FDC window at C000. The `startup` list of the machine file runs the PROM. RDOS reads the boot track through the WD FD1793 controller and loads the CDOS cold loader. The loader reads `CDOS.COM` into memory and jumps to it.

CDOS then moves itself to the top of RAM and switches the PROM out with `OUT 40H`. The memory board hides the RAM under the PROM on reads only. The boot reads the PROM, and a write reaches the RAM under it. Because of this, CDOS can put running code in that RAM.

Two authentic settings

- **4 MHz.** A real Cromemco ran at 4 MHz. The double-density read loop of RDOS only keeps up with a 500 kbit/s diskette at that speed. At 2 MHz, the controller reports Lost Data and the boot fails. Do not change `clock_hz = 4000000`.
- **The mixed-density diskette.** `CDOS258-8IN-DSDD.DSK` is an 8" double-sided image. Track 0 is single density, because the boot PROM reads only single density. The other tracks are double density. The 16FDC finds the format from the image. You do not set it.

Console speed

The console runs at full speed. Output appears as fast as CDOS prints it, even with the 300-baud modem setting. To get the speed of a 300-baud modem console, set the rate on the console. A `DIR` then takes many seconds.

```
[board.unit.tty]
connect = "console"
rate    = "real"    # pace the line in real time at the programmed baud
```

`rate = "full"` is the default. It does not pace the emulated line. `rate = "real"` does. When a real serial port is on the other end, that port keeps its own timing. The `rate` setting changes only the timing of the emulated console.

The disk image

The repository tracks `CDOS258-8IN-DSDD.DSK`, and the machine file mounts it read/write in drive A, as a real machine does. Booting, `DIR` and `STAT` only read it. Before you test writes, copy the image, or add

`writeprotect = true` to the drive in the machine file. In a clone, `git checkout` restores a changed image. A downloaded copy has no such way to restore it.